

EMILIA GovGuard

Pre-execution payment integrity for public funds

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The control gap

Most fraud against public funds happens *inside valid sessions*. The actor is authenticated. The system records a successful action. The control surface verifies *who is logged in* — it does not produce a tamper-evident, cryptographically bound record proving that **this exact change** was authorized by **the right named human** under **the right policy** at **the exact moment** it executed.

That gap is where benefit-routing fraud, vendor-bank-account redirection, and operator-override misuse succeed.

What GovGuard does

GovGuard is a pre-execution control layer. It sits between the case-system save and the payment system, intercepting high-risk actions before they commit:

- **Verifies actor identity.** Cryptographically — not just session-level.
- **Verifies authority chain.** Complete delegation path from root authority to the acting principal.
- **Binds exact action context.** Specific operation, target, parameters, environmental conditions — pinned to a one-time cryptographic ceremony.
- **Pins policy version and hash.** Immutable reference to the exact policy that authorized this action.
- **Requires named human signoff.** Where policy demands it, a specific accountable human assumes irrevocable ownership of the outcome — not a role, a person.
- **Issues a tamper-evident receipt.** Ed25519-signed, Merkle-anchored, verifiable offline by anyone.

What auditors get

Every controlled action produces an evidence packet:

- decision record · policy version + hash · actor identity source · authority chain · approval sequence · signoff trace · nonce, expiry, and one-time consumption proof · execution reference · final outcome

Live example a buyer can verify themselves: <https://emiliaprotocol.ai/r/example>

Anyone can install `@emilia-protocol/verify` and re-check the signature without trusting our infrastructure.

30-day pilot scope

Phase	Duration	Activity
Week 0	5 days	Workflow selected, source systems identified, risk thresholds set, approval rules configured, success metrics agreed
Week 1	7 days	Shadow mode — GovGuard observes, never blocks. Daily report: high-risk actions, missing-authority signals, after-hours actions, new-destination changes
Week 2	7 days	Shadow + receipt mode — every high-risk action gets a Trust Receipt preview
Week 3	7 days	Limited enforcement on one class (e.g., vendor bank-account changes $\geq$ \$10K)
Week 4	4 days	Final report — high-risk actions found, approvals required, evidence packets generated, expansion proposal

Pilot fee: \$25,000 – \$75,000 depending on integration depth. **Annual license** (post-pilot): \$150,000 – \$500,000 or per-action pricing (\$0.50–\$5/action), whichever the agency prefers.

Open standard, no lock-in

- **License:** Apache 2.0 — agency can self-host, fork, or independently verify.
- **Formal verification:** 26 TLA+ theorems verified, 32 Alloy facts + 15 assertions, both run in CI.
- **Compliance mappings:** NIST AI RMF (38 subcategories), EU AI Act Articles 9–15 + 26.
- **Open-source verifier:** `npm install @emilia-protocol/verify` — zero-dependency, works offline.

Contact

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